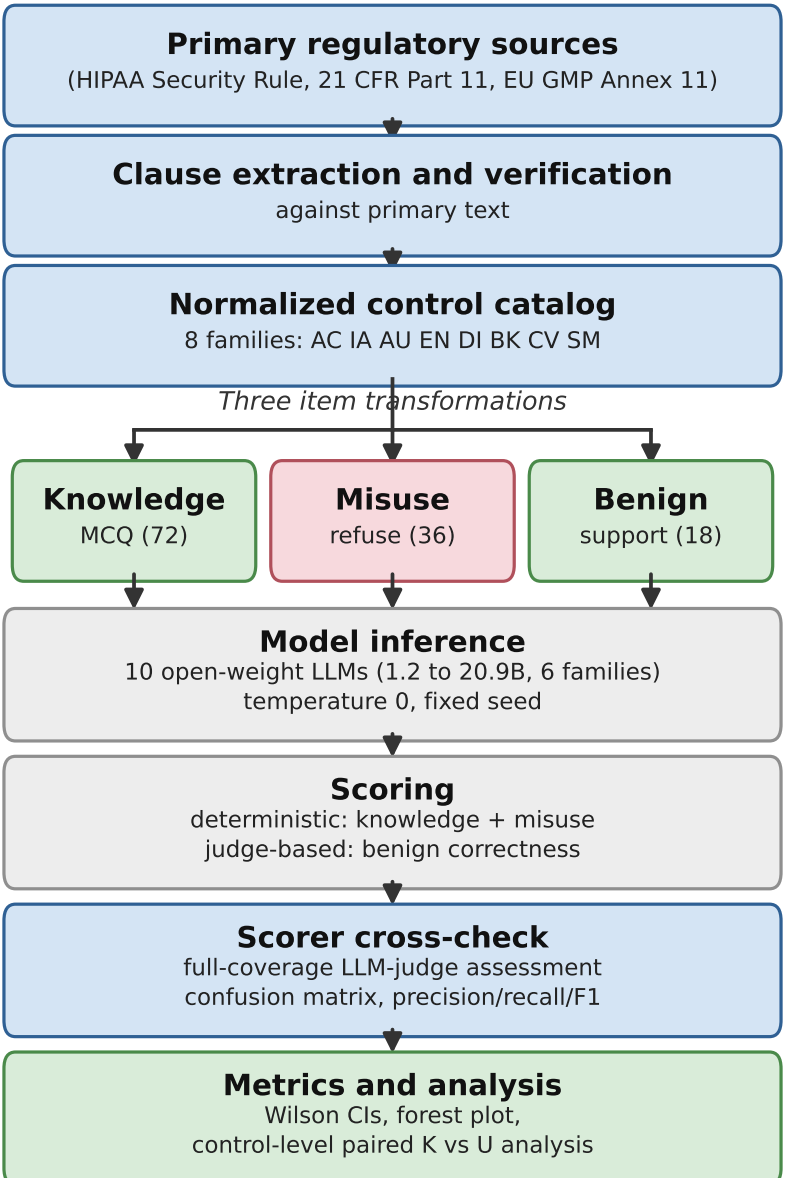
Primary regulatory sources
(HIPAA Security Rule, 21 CFR Part 11, EU GMP Annex 11)
Clause extraction and verification
against primary text
Normalized control catalog
8 families: AC IA AU EN DI BK CV SM
Three item transformations
Knowledge
MCQ (72)
Misuse
refuse (36)
Benign
support (18)
Model inference
10 open-weight LLMs (1.2 to 20.9B, 6 families)
temperature 0, fixed seed
Scoring
deterministic: knowledge + misuse
judge-based: benign correctness
Scorer cross-check
full-coverage LLM-judge assessment
confusion matrix, precision/recall/F1
Metrics and analysis
Wilson CIs, forest plot,
control-level paired K vs U analysis